

MSDI Calculator

EIA Energy Price API

Methodology

Technical Methodology Memorandum

Prepared for internal model documentation and technical review

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Model scope: Manure Sub-Surface Drip Irrigation (mSDI-e) economic screening calculator

Document status: Methodology documentation for EIA Energy Price API integration in costs section of user form.

Purpose and Scope

This technical memorandum documents the methodology used to incorporate U.S. Energy Information Administration (EIA) energy price defaults into the MSDI Calculator. The integration provides optional user-initiated defaults for fuel and electricity costs on the Costs page of the application. The EIA functionality is intended to reduce data-entry burden while preserving user control over final model inputs.

The methodology describes the role of the EIA lookup table, the client-side API utility, the selected EIA data products, unit conversions, state and diesel-region mapping, user-interface attribution, QA/QC expectations, and change-management requirements. The API defaults are not treated as fixed model coefficients. They are external, time-varying defaults that users may accept or edit before calculation.

Terminology and Units

Term	Definition in this methodology	Primary artifact or unit
EIA	U.S. Energy Information Administration, the public data source used for fuel and electricity price defaults.	External public data source
EIA API key	Registered key used by the application build to request EIA Open Data API records.	VITE_EIA_API_KEY
Commercial electricity price	Latest available monthly state-level commercial electricity price returned by EIA.	cents/kWh from EIA; \$/kWh in app
ULSD diesel price	Latest weekly ultra-low sulfur diesel retail price returned by EIA for the mapped diesel region.	\$/gallon
EIA lookup table	Application lookup table mapping app states to EIA electricity state codes and EIA diesel regions.	lookup-eia.json
User-editable default	A value populated from EIA only after user action and editable before model submission.	form input value
Attribution message	User-facing source and period note displayed after EIA data retrieval.	text status message

EIA Data Sources and Selected Series

The integration uses two EIA data products that align with the Costs page inputs. Fuel cost defaults are based on weekly retail ultra-low sulfur diesel price data. Electricity cost defaults are based on monthly state-level commercial electricity price data. These values are used only when the user clicks the corresponding Use EIA Data button.

App input	EIA data selection	Frequency	Geographic basis	Unit handling
Fuel cost (\$/diesel gallon)	No. 2 Diesel Ultra Low Sulfur (0-15 ppm) Retail Prices	Weekly	Mapped EIA diesel region	Returned as \$/gallon; no conversion required
Electricity cost (\$/kWh)	Retail Sales of Electricity, commercial sector price	Monthly	Mapped EIA state code	Returned as cents/kWh; divided by 100 before population

EIA Lookup Table and Geographic Mapping

The EIA integration uses a workbook-generated lookup table rather than hard-coded mappings in the API utility. The EIA_LU worksheet is maintained in the source workbook and regenerated into src/data/lookups/lookup-eia.json using the existing lookup generation workflow. Each row maps the application state identifier to the EIA electricity state code and the EIA diesel pricing region used by the API calls.

This lookup-driven approach is consistent with the broader MSDI Calculator architecture. It allows state-to-EIA mappings to be reviewed in tabular form, regenerated through the standard lookup script, and version-controlled alongside other model lookup updates.

Lookup field	Role in methodology	Implementation implication
app_state	Stores the application-facing state name or label.	Used for review and documentation.
key / value	Stores the normalized application state identifier used by the form.	Used to retrieve the correct EIA mapping for the selected state.
eia_electricity	Stores the two-letter EIA state code used for electricity price requests.	Example: California -> CA.
eia_diesel	Stores the EIA duoarea code used for diesel price requests.	Examples: SCA, R5XCA, R50, R40, R30, R20, R10.
fips	Stores the state FIPS value for consistency with the state lookup architecture.	Supports QA/QC against state lookup records.

Application Integration Workflow

The EIA price default workflow is initiated by the user on the Costs page. The application reads the selected state from the form state, retrieves the matching EIA lookup record, sends the appropriate API request, and populates the relevant input only after a successful response. The populated value remains editable by the user.

Workflow step	Implemented behavior
Read selected state	The EIA default component reads the state previously selected on the Farm Location page through react-hook-form.
Resolve EIA mapping	src/lib/eia.ts retrieves the matching row from lookup-eia.json and obtains the EIA electricity code or diesel region.
Request EIA data	The API utility requests the latest available record using descending period sort and length=1.
Convert units where needed	Commercial electricity prices are divided by 100 to convert cents/kWh to \$/kWh. Diesel prices remain in \$/gallon.
Populate form value	setValue updates fuel_cost_per_gallon or electricity_cost_per_kwh and triggers validation.
Display attribution	The user interface displays EIA source, period, and editability messages after retrieval.

Unit Conversion and Value Handling

The diesel endpoint returns retail diesel price values in dollars per gallon, matching the fuel-cost unit expected by the application. No conversion is applied to diesel values. The electricity endpoint returns commercial electricity prices in cents per kilowatt-hour. The application divides the EIA electricity value by 100 before populating the electricity cost field so that the value is expressed as dollars per kilowatt-hour.

EIA value example	Conversion	Application value
Diesel: 4.458 \$/GAL	No conversion	4.458 \$/gallon
Electricity: 22.4792 cents/kWh	22.4792 / 100	0.224792 \$/kWh

User Interface and Source Attribution

The EIA defaults are presented as optional, field-specific assistance directly below the corresponding fuel and electricity inputs. Button labels are intentionally simple, using Use EIA Data, while nearby explanatory text describes the EIA data product, geography, frequency, and unit handling. This presentation keeps the source transparent and makes clear that the populated values are defaults rather than farm-specific billing records.

Status messages identify the U.S. Energy Information Administration as the source, provide the EIA reporting period used for the populated value, and state that the value may be edited. The application does not use the EIA logo or imply EIA endorsement.

QA/QC and Validation Workflow

QA/QC check	Purpose
Verify EIA_LU generation	Confirm lookup-eia.json is regenerated from the workbook and committed with the source workbook.
Check state mapping coverage	Confirm every application state has an eia_electricity and eia_diesel value.
Validate API responses	Test representative states, including California and non-California states, for both diesel and electricity retrieval.
Validate units	Confirm diesel is populated as \$/gallon and electricity is converted from cents/kWh to \$/kWh.
Validate source messaging	Confirm EIA attribution and reporting period are visible after retrieval.
Validate user editability	Confirm populated values can still be changed before submission and downstream calculation.
Validate production build	Confirm the GitHub Actions build step receives VITE_EIA_API_KEY and the deployed site can retrieve data.

Assumptions and Limitations

- EIA values are optional defaults and do not replace farm-specific fuel invoices, utility bills, or rate schedules.
- Commercial electricity price data are used as the most appropriate default sector for farm electricity costs unless a user has more specific tariff information.
- Electricity data are monthly and may lag the current date because EIA monthly releases are not real-time billing data.
- Diesel data are weekly and regional or sub-regional, not farm-specific delivered fuel prices.
- The EIA API key is provided to the Vite build through a GitHub Actions repository secret for production deployment.
- Because the integration is client-side, API-request behavior depends on EIA API availability and browser network access.
- The user remains responsible for reviewing and editing populated defaults before submitting the calculator.

Processing Workflow Summary

1. User selects a state on the Farm Location page.
2. User reaches the Costs page and reviews fuel and electricity inputs.
3. User clicks Use EIA Data for diesel and/or electricity.
4. The EIA default component reads the selected state from form state.
5. src/lib/eia.ts retrieves the matching lookup-eia.json record.
6. The utility calls the appropriate EIA API endpoint and requests the latest available record.
7. The application converts electricity from cents/kWh to \$/kWh and leaves diesel in \$/gallon.
8. The corresponding input is populated and validated.
9. The interface displays EIA source attribution, reporting period, and editability language.
10. The user may keep, revise, or replace the populated value before model submission.

Implementation Traceability

Methodology area	Primary implementation file or artifact	Traceability note
EIA mapping source	src/data/raw/msdi_lookup_tables.xlsx; EIA_LU worksheet	Editable source of truth for EIA state and diesel-region mappings.
Generated EIA lookup	src/data/lookups/lookup-eia.json	Generated lookup consumed by the EIA API utility.
Lookup generation	pnpm run generate:lookups	Regenerates lookup-eia.json from the source workbook.
API utility	src/lib/eia.ts	Builds EIA API requests, reads lookup mappings, converts electricity units, and returns value/period/source objects.
User interface controls	src/components/forms/eia-defaults.tsx	Renders field-specific Use EIA Data buttons, helper text, status messages, and source attribution.
Form placement	src/components/forms/multi-step-form.tsx	Places EIA default controls beneath the fuel and electricity inputs on the Costs page.
Production configuration	.github/workflows/azure-static-web-apps-lively-ocean-037806a10.yml	Provides VITE_EIA_API_KEY to the Vite build through GitHub Actions secrets.
Change management	CHANGELOG.md	Documents EIA integration, lookup mapping, workflow configuration, and UI changes.

References and Source Links

U.S. Energy Information Administration. EIA Open Data API documentation and terms of service. Public data API documentation, accessed 2026.

EIA electricity retail sales API. Monthly state-level commercial electricity price data, returned in cents per kilowatt-hour.

EIA petroleum prices API. Weekly No. 2 Diesel Ultra Low Sulfur (0-15 ppm) retail price data, returned in dollars per gallon.

MSDI Calculator EIA lookup worksheet. EIA_LU in src/data/raw/msdi_lookup_tables.xlsx. Internal project lookup workbook, 2026.

MSDI Calculator EIA utility. src/lib/eia.ts. Internal TypeScript implementation, 2026.

MSDI Calculator EIA defaults component. src/components/forms/eia-defaults.tsx. Internal TypeScript implementation, 2026.

MSDI Calculator Change Log. CHANGELOG.md. Internal project change management record, 2026.